NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED **MATHEMATICS** **M.M. 78**

REVISION ASSINGMENT # 18 (QUADRATIC EQUATION + SEQUENCE SERIES) TIME : 70 MIN.

SECTION–I(i)

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If α,β are the integral roots of the quadratic equation $x^2 - \alpha\beta x + \alpha^2 + \beta = 0$, then number of possible different quadratic equation(s) is-

(A) 0 (B) 1 (C) 2 (D) 4

2. If roots of the quadratic equation $(x-\alpha)(x-\beta)+(x-\gamma)(x-\delta)=0$ are m & n respectively, then roots of the equation $2(x-m)(x-n)-(x-\alpha)(x-\beta)=0$ will be

(A) $\frac{\alpha+m+\gamma}{2}, \frac{\beta+n+\delta}{2}$ (B) γ, δ

(C) $\alpha - m, \beta - n$ (D) $m-\alpha+\gamma, n-\beta+\delta$

3. If $a \in (0, 10)$ such that roots of the equation $x^2 - 4x - a^2 + 2a + 3 = 0$ lies between the roots of the equation $x^2 - 4x + 4 - a^2 = 0$ then number of integral values of 'a' are -

(A) 5 (B) 6 (C) 8 (D) 9

4. If α & β are real roots of $(\log_3 x)^2 - 6\log_3 x + k = 0$, $k \in R$, then minimum integral value of $(\alpha+\beta)$ is-

(A) 53 (B) 54 (C) 55 (D) 56

5. Let $a_1, a_2, \ldots\ldots a_n$ be a sequence such that $a_1 = 2$ and $a_n - a_{n-1} = n^2 \ \forall\ n \geq 2$. If $a_k = 205$ then k is equal to-

(A) 5 (B) 8 (C) 11 (D) 15

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6. If common ratio of a geometric progression is positive then maximum value of the ratio of second term to the sum of first 3 terms equals -

(A) 2 (B) $\frac{1}{3}$ (C) 3 (D) $\frac{1}{2}$

7. The minimum value of $\frac{12}{x}+\frac{18}{y}+xy$, where x and y are positive real numbers, is -

(A) $12\sqrt{3}$ (B) $24\sqrt{3}$ (C) 18 (D) infinity

8. Let A_i, i = 1, 2,....6 represents 6 AMs
G_i, i = 1,2,3,.....6 represents 6 GMs
H_i, i = 1, 2, ...,6 represents 6 HMs between two numbers 2 & 72

and G represents single GM between 2 & 72, then the value of $\frac{\left(\prod_{i=1}^{6} G_i\right)\left(\sum_{i=1}^{6}\frac{1}{H_i}\right)}{\left(\sum_{i=1}^{6} A_i\right)}$ is -

(A) G (B) G^2 (C) G^4 (D) G^6

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Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. If equations $ax^2-(a+b)x+b=0$ & $bx^2+(b-c)x-c=0$ has exactly one root in common $\{a, b, c \neq 0\}$, then which of the following can be correct -

(A) $b^2=ac$ (B) $-a=b\neq c$ (C) $b=a\neq c$ (D) $-a\neq b=c$

10. Let $\alpha, \beta, \gamma, \delta$ be the roots of $x^4-x^3-x^2-1=0$. Also consider $p(x)=x^6-x^5-x^3-x^2-x$. Then the value of $p(\alpha)+p(\beta)+p(\gamma)+p(\delta)$ cannot be

(A) 4 (B) 5 (C) 6 (D) 7

11. Two consecutive numbers k, k +1 are removed from natural numbers 1,2,3,.....n such that arithmetic mean reduces by 1, then (n – k) is equal to -

(A) Prime number (B) Neither prime nor composite number

(C) Odd number (D) Even number

12. If positive numbers x, y, z are in A.P., then the minimum value of $\frac{x+y}{2y-x}+\frac{y+z}{2y-z}$ is less than

(A) 2 (B) 6 (C) 8 (D) –1

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 13 to 14

Let $f(x)=2x^2-2(2\lambda+1)x+\lambda(\lambda+1)=0$, where λ is a parameter.

On the basis of above information, answer the following questions :

13. If both roots of equation $f(x)=0$ belong to $(-1,1)$, then greatest integral value of $|10\lambda|$ is -

(A) 2 (B) 4 (C) 5 (D) 9

14. If equation $ax^2+bx+c=0$, where a, b & c are sides of any triangle and equation $f(x)=0$ having both roots common, then set of possible values of λ is -

(A) $(-\infty,-2)$ (B) $(-5, 0)$ (C) $(-5,-4)$ (D) $(4,\infty)$

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SECTION–II (iii)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. Let α,β are the roots of the equation $x^2 + ax + b = 0$ and γ,δ are the roots of $x^2 - ax + b - 2 = 0$. If $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{5}{12}$ & $\alpha\beta\gamma\delta = 24$, then the value of 'a' is

2. The only integer which does not lie in the range of the function $f(x) = \frac{x^2-3x+2}{x^2+x-6}$ is

3. If $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots\ldots$ upto n terms is equal to 1575, then the value of n will be

4. A number consists of three digits which are in G.P. The sum of the right hand and left hand digit exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digit. Find the numbers.

5. $1+\frac{(1+2)(1+3)}{2^2\left(1^3+2^3\right)}+\frac{(1+2+3)(1+3+5)}{3^2\left(1^3+2^3+3^3\right)}+\frac{(1+2+3+4)(1+3+5+7)}{4^2\left(1^3+2^3+3^3+4^3\right)}+\ldots\ldots$ upto10 terms $=\frac{a}{b}$ (where a & b are relatively prime), then a – b is equal to

6. If $(1 + 3 + 5 + \ldots\ldots + a) + (1 + 3 + 5 + \ldots\ldots + b) = (1 + 3 + 5 \ldots\ldots + c)$, such that (i) $a + b + c = 57$ and (ii) $a < 10$, then the value of $(2b - 2c + a)$ is equal to

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SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		Column-II
(A)	If p^{th} , q^{th} and r^{th} terms of a G.P. (common ratio $\neq 1$) are in G.P. then p,q,r will be in	(P)	Arithmetic progression
(B)	If a,b,c,d are non-zero real numbers such that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \leq (ab + bc + cd)^2$ and $c(a - b) = a(b - c)$, then a,b,c,d will be in	(Q)	Geometric progression
(C)	If a,b,c,d are four different positive numbers in G.P. then $\frac{a(2\log a+\log b)}{b}, \frac{b(2\log b+\log c)}{c}, \frac{c(2\log c+\log d)}{d}$ will be in	(R)	Harmonic progression
(D)	If a_1,a_2,a_3 are in A.P., a_2,a_3,a_4 are in G.P. and a_3,a_4,a_5 are in H.P. then a_1,a_3,a_5 are always in	(S)	Not a harmonic progression
		(T)	A progression, whose all the terms are identical

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REVISION ASSINGMENT # 17 **DE + AUC** **MATHEMATICS**

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	C	D	D	D	B	C	C	D	A,B,D	A,B
	Q.	11	12	13							
	A.	B,C,D	A,B,C	B,C,D							
SECTION-II	Q.	1	2	3	4	5					
	A.	4.00	1.00	2.00	7.00	12.00					

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